"FLOOD ALARM USING ARDUINO"

A MINI PROJECT REPORT

Submitted

In the partial fulfilment of the requirements for
the award of the degree of

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In

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By

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DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING
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CERTIFICATE

This is to certify that Mini Project report entitled "**FLOOD ALARM USING ARDUINO**" embodies the original work done by **T.Dharani** bearing the Roll Number **B17EC145** studying VI Semester in partial fulfilment of the requirement for the award of degree of the **Bachelor of Technology** in **Electronics & Communication Engineering** from **KAKATIYA INSTITUTE OF TECHNOLOGY & SCIENCE, WARANGAL** during the academic year 2019-2020.

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ABSTRACT

Flood is an unavoidable disaster which causes severe damage to properties and lives. For this reason, researchers created a flood detection system to monitor rising water in residential areas. The early warning systems for flood management have been developed rapidly with growth of technologies. These warning systems help to alert the people earlier.

Many proposed system used sophisticated techniques to alert flood such as Short Message Service (SMS) via Global System for Mobile communications (GSM), however, they are still complex to program and interface, in addition to their high cost. This paper proposed a simple, portable and low cost of early warning system using Arduino board, which is used to control the whole system and GSM shields to send and receive data. The proposed model determines the water levels using ranging sensors. Then it analyses the collected data and determine the type of danger present. The concept of portable sensor for this project enables the sensor to be deployed only during the needed period.

Contents

1. INTRODUCTION

Floods impact on both individuals and communities, and have social, economic, and environmental consequences. The consequences of floods, both negative and positive, vary greatly depending on the location and extent of flooding, and the vulnerability and value of the natural and constructed environments they affect.

Floods are caused by many different incidents including natural disasters as well as manmade causes. Natural disasters include mudflows, flash floods, and heavy rains. Most manmade floods occur with a pipe bursting, plumbing problems, or when a malfunctioning water heater is present. 70% of floods are actually due to water heaters bursting or leaking.

Flood alarms are essential to prevent damage when a disaster arises. These damages are not only a nuisance, but very costly. So it is essential to repair damages as quickly as possible.

2. BLOCK DIAGRAM

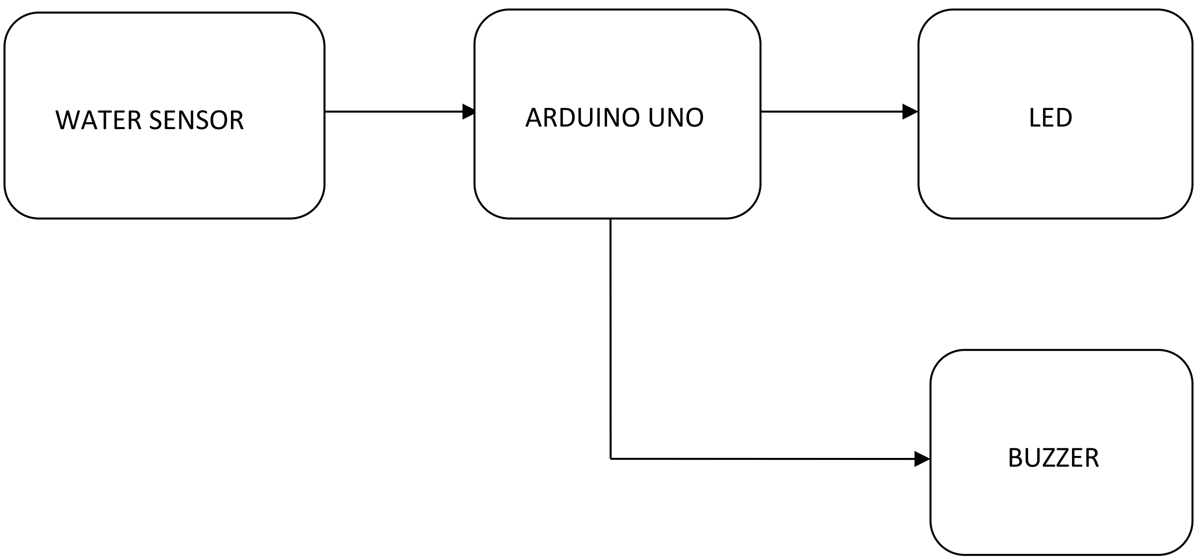

Fig 2.1.1 block diagram of flood alarm using arduino

3. COMPONENTS DESCRIPTION

3.1 Arduino UNO:

The Arduino Uno is one kind of microcontroller board based on ATmega328, and Uno is an Italian term which means one. Arduino Uno is named for marking the upcoming release of microcontroller board namely Arduino Uno board 1.0. This board includes digital I/O pins-14, a power jack, analog i/ps-6, ceramic resonator-A16 MHz, a USB connection, an RST button, and an ICSP header. All these can support the microcontroller for further operation by connecting this board to the computer. The power supply of this board can be done with the help of an AC to DC adapter, a USB cable otherwise a battery.

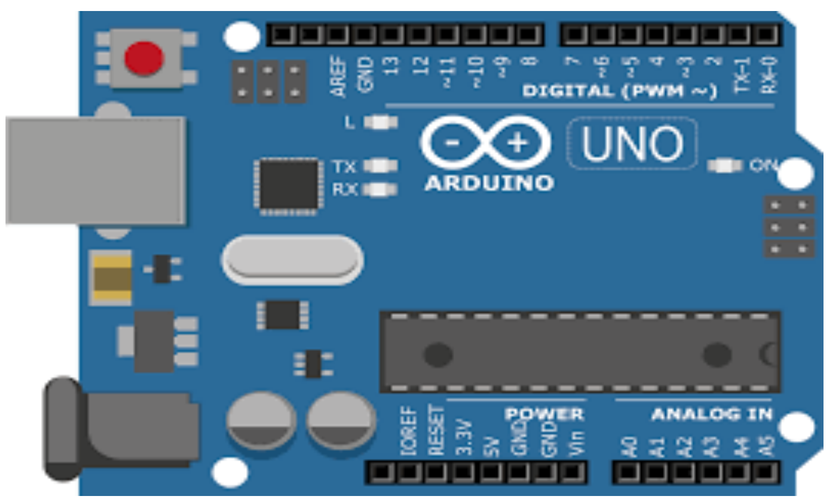

Fig 3.1.1 Arduino Uno

Features of Arduino Uno Board

The **features of Arduino Uno ATmega328** includes the following.

The operating voltage is 5V
- The recommended input voltage will range from 7v to 12V
- The input voltage ranges from 6v to 20V
- Digital input/output pins are 14
- Analog i/p pins are 6
- DC Current for each input/output pin is 40 mA

- Flash Memory is 32 KB
- SRAM is 2 KB
- EEPROM is 1 KB
- CLK Speed is 16 MHz

Arduino Uno Pin Diagram

The Arduino Uno board can be built with power pins, analog pins, ATmegs328, ICSP header, Reset button, power LED, digital pins, test led 13, TX/RX pins, USB interface, an external power supply. The Arduino Uno board description is discussed below.

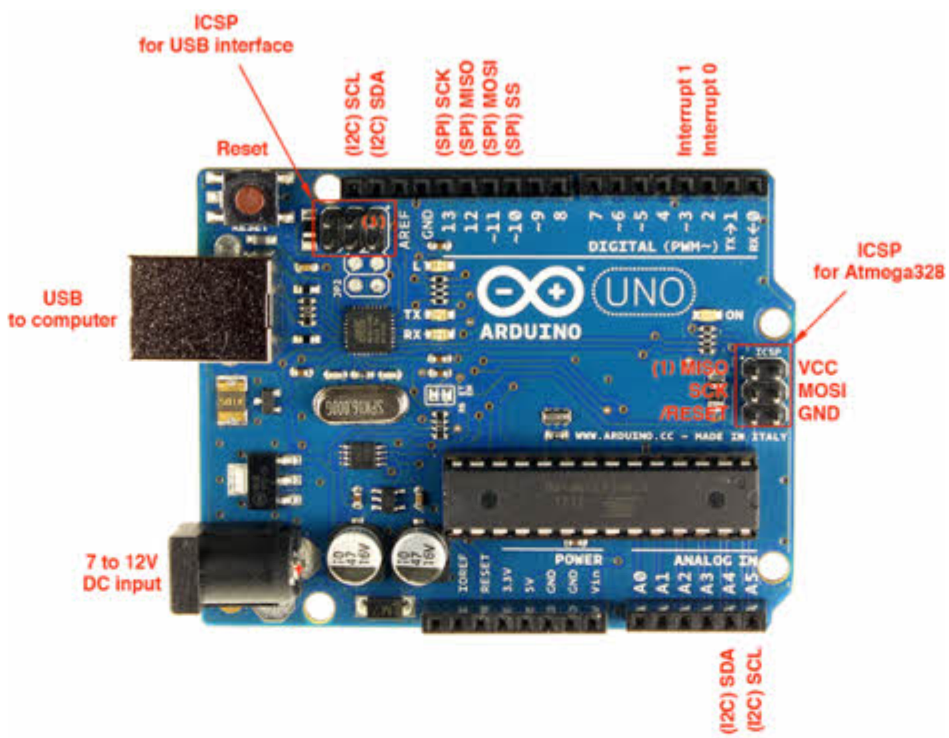

Fig.3.1.2 Arduino pin configuaration

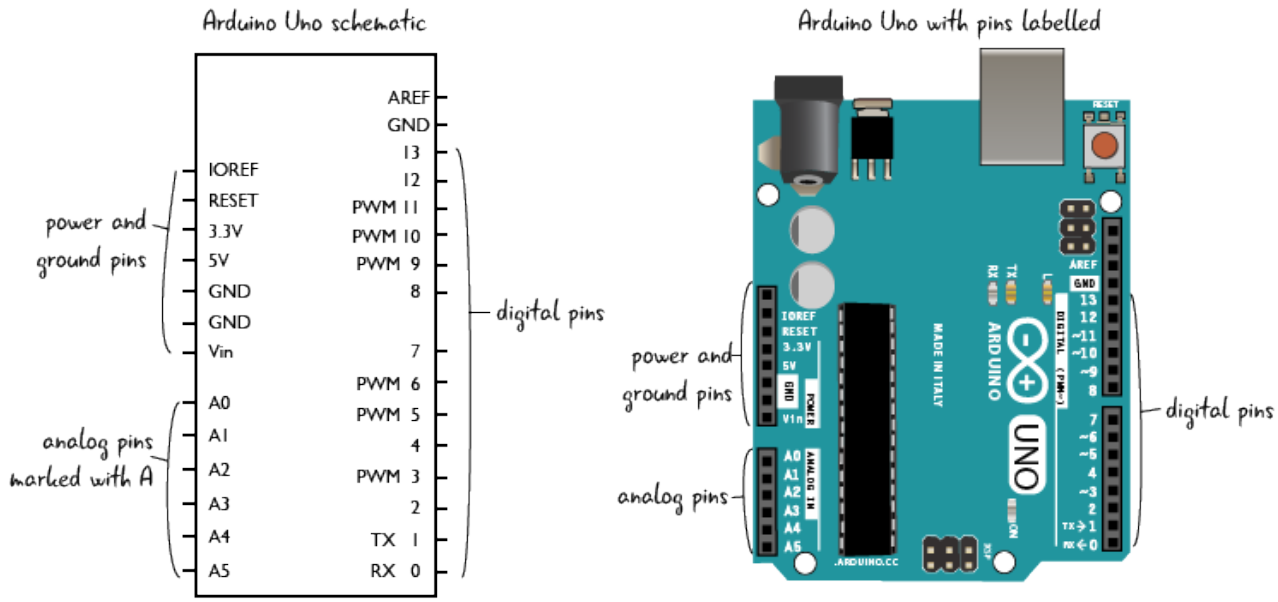

Fig.3.1.3 Arduino UNO pin diagram

Power Supply:

The **Arduino Uno power supply** can be done with the help of a USB cable or an external power supply. The external power supplies mainly include AC to DC adapter otherwise a battery. The adapter can be connected to the Arduino Uno by plugging into the power jack of the Arduino board. Similarly, the battery leads can be connected to the Vin pin and the GND pin of the POWER connector. The suggested voltage range will be 7 volts to 12 volts.

Input & Output:

The 14 digital pins on the Arduino Uno can be used as input & output with the help of the functions like pinMode(), digitalWrite(), & Digital Read().

Pin1 (TX) & Pin0 (RX) (Serial):

This pin is used to transmit & receive TTL serial data, and these are connected to the ATmega8U2 USB to TTL Serial chip equivalent pins.

Pin 2 & Pin 3 (External Interrupts):

External pins can be connected to activate an interrupt over a low value, change in value.

Pins 3, 5, 6, 9, 10, & 11 (PWM):
This pin gives 8-bit PWM o/p by the function of analogWrite().

SPI Pins (Pin-10 (SS), Pin-11 (MOSI), Pin-12 (MISO), Pin-13 (SCK):
These pins maintain SPI-communication, even though offered by the fundamental hardware, is not presently included within the Arduino language.

Pin-13(LED):
The inbuilt LED can be connected to pin-13 (digital pin). As the HIGH-value pin, the light emitting diode is activated, whenever the pin is LOW.

Pin-4 (SDA) & Pin-5 (SCL) (I2C):
It supports TWI-communication with the help of the Wire library.

AREF (Reference Voltage):
The reference voltage is for the analog i/ps with analogReference().

Reset Pin:
This pin is used for reset (RST) the microcontroller.

Memory:
The memory of this Atmega328 Arduino microcontroller includes flash memory-32 KB for storing code, SRAM-2 KB EEPROM-1 KB.

Communication
The Arduino Uno ATmega328 offers UART TTL-serial communication, and it is accessible on digital pins like TX (1) and RX (0). The software of an Arduino has a serial monitor that permits easy data. There are two LEDs on the board like RX & TX which will blink whenever data is being broadcasted through the USB.

3.2 Water level sensor:

Water sensor is designed for water detection, which can be widely used in sensing rainfall, water level, and even liquid leakage. Connecting a water sensor to an arduino is a great way to detect a leak, spill, food, rain etc. Arduino water level sensor comes in many sizes and with many names. But universally they all are composed of same electronic component and works on same principle. All the sensors are analog. Means they provide analog output. Analog output is in range from 0 to 5 volts. We need to read this analog voltage using ADC (analog to digital converter) and convert it to equivalent digital value for future computations.

- Arduino water sensor can be used to check the leakage of water or viscous substance from pipes.
- Water detector can be used to check spil of water from tank.
- Water sensor can measure the salinity or moisture water content in soil.
- Flood and rain situation can be monitored.
- Quantity or level of water in tank can easily be monitored using arduino water level sensor.
- Water sensor can be utilized to monitor and water the garden.

Water sensor brick is designed for water detection, which can be widely used in sensing the rainfall, water level, even the liquid leakage. The brick is mainly comprised of 3 parts: An Electronic brick connector, a 1 MΩ resistor, and several lines of bare conducting wires.

This sensor works by having a series of exposed traces connected to ground and interlaced between the grounded traces are the sense traces. The sensor traces have a weak pull-up resistor of 1 MΩ. The resistor will pull the sensor trace value high until a drop of water shorts the sensor trace to the grounded trace. Believe it or not this circuit will work with the digital I/O pins of your Arduino or you can use it with the analog pins to detect the amount of water induced contact between the grounded and sensor traces.

It can judge the water level through with a series of exposed parallel wires stitch to measure the water droplet/water size. It can easily change the water size to analog signal, and output analog value can directly be used in the program function, then to achieve the function of water level alarm. It has low power consumption, and high sensitivity, which are the biggest characteristics of this module.

Water sensor pin configuration:

The water level sensor is super easy to use and only has 3 pins to connect.

S(signal) pin is an analog output that will be connected to one of the analog inputs on your Arduino.
+Vcc pin supplies power for the sensor. It is recommended to power the sensor with between 3.3V – 5V. Please note that the analog output will vary depending on what voltage is provided for the sensor.
-(GND) is a ground connection.

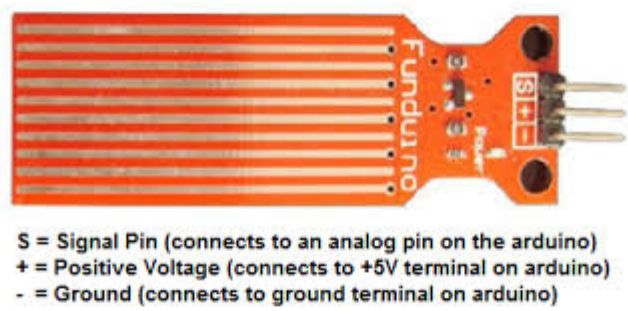

Fig. 3.2.1 Pin configuration of water level sensor

Features :

1、Working voltage: 5V/3.3V
2、Working Current: <20ma
3、Interface: Analog
4、Width of detection: 40mm×16mm
5、Working Temperature: 10℃~30℃
6、Weight: 3g
7、Size: 65mm×20mm×8mm
8、Arduino compatible interface

9、Low power consumption
10、High sensitivity
11、Output voltage signal: 0~4.2V
12、Production Technology: FR4 double-sided spray tin

Interfacing water level sensor with arduino uno :

We can interface water detector with arduino in digital as well as analog form. Let's start with the analog interface. Arduino will read the analog output of the water sensor and display it on serial monitor of arduino.

First you need to supply power to the sensor. For that you can connect the + (VCC) pin on the module to 5V on the Arduino and – (GND) pin to ground.

Connect the analog output of arduino water sensor to arduino analog channel 1. The code of the project is simple. In **setup()** function of arduino code serial monitor of arduino is initialized at 9600 bps. The instruction **Serial.begin(9600)** is starting the serial monitor. In the loop function i am reading the analog channel 1 of ADC. The instruction **int sensor=analogRead(A1)** reads the analog channel and stores the analog value in variable **sensor**. Next instruction **Serial.println(sensor)** prints the water sensor analog reading on serial monitor of arduino.

Applications

Water sensors find applications in nuclear power plants, automobiles for measuring the amount of gasoline left in the fuel tank, engine oil, cooling water, and brake/power steering fluid.

Industrial applications of water level sensors include water level sensing in transport and storage tanks and water treatment tanks. Wess Global Inc. manufactures different types of level instruments for their use in municipal areas and also in the food and beverage industry.

3.3. LED

A light-emitting diode (LED) is a semiconductor device that emits visible light when an electric current passes through it. The light is not particularly bright, but in most LEDs it is monochromatic, occurring at a single wavelength. The output from an LED can range from red (at a wavelength of approximately 700 nanometers) to blue-violet (about 400 nanometers).

The light emitting diode simply, we know as a diode. When the diode is forward biased, then the electrons & holes are moving fast across the junction and they are combining constantly, removing one another out. Soon after the electrons are moving from the n-type to the p-type silicon, it combines with the holes, then it disappears. Hence it makes the complete atom & more stable and it gives the little burst of energy in the form of a tiny packet or photon of light.

Light Emitting Diodes or simply LED´s, are among the most widely used of all the different types of semiconductor diodes available today and are commonly used in TV's and colour displays. They are the most visible type of diode, that emit a fairly narrow bandwidth of either visible light at different coloured wavelengths, invisible infra-red light for remote controls or laser type light when a forward current is passed through them. The "Light Emitting Diode" or LED as it is more commonly called, is basically just a specialised type of diode as they have very similar electrical characteristics to a PN junction diode. This means that an LED will pass current in its forward direction but block the flow of current in the reverse direction. Light emitting diodes are made from a very thin layer of fairly heavily doped semiconductor material and depending on the semiconductor material used and the amount of doping, when forward biased an LED will emit a coloured light at a particular spectral wavelength.

When the diode is forward biased, electrons from the semiconductors conduction band recombine with holes from the valence band releasing sufficient energy to produce photons which emit a monochromatic (single colour) of light. Because of this thin layer a reasonable number of these photons can leave the junction and radiate away producing a coloured light output.

LED Construction :

Then we can say that when operated in a forward biased direction Light Emitting Diodes are semiconductor devices that convert electrical energy into light energy.The construction of a Light Emitting Diode is very different from that of a normal signal diode. The PN junction of an LED is surrounded by a transparent, hard plastic epoxy resin hemispherical shaped shell or body which protects the LED from both vibration and shock.

Surprisingly, an LED junction does not actually emit that much light so the epoxy resin body is constructed in such a way that the photons of light emitted by the junction are reflected away from the surrounding substrate base to which the diode is attached and are focused upwards through the domed top of the LED, which itself acts like a lens concentrating the amount of light. This is why the emitted light appears to be brightest at the top of the LED. However, not all LEDs are made with a hemispherical shaped dome for their epoxy shell. Some indication LEDs have a rectangular or cylindrical shaped construction that has a flat surface on top or their body is shaped into a bar or arrow. Generally, all LED's are manufactured with two legs protruding from the bottom of the body. Also, nearly all modern light emitting diodes have their cathode, (–) terminal identified by either a notch or flat spot on the body or by the cathode lead being shorter than the other as the anode (+) lead is longer than the cathode (k).

Unlike normal incandescent lamps and bulbs which generate large amounts of heat when illuminated, the light emitting diode produces a "cold" generation of light which leads to high efficiencies than the normal "light bulb" because most of the generated energy radiates away within the visible spectrum. Because LEDs are solid-state devices, they can be extremely small and durable and provide much longer lamp life than normal light sources.

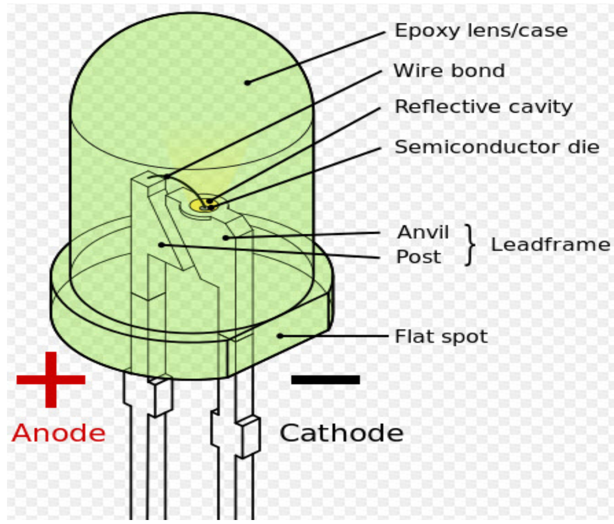

Fig 3.3.1 Light Emitting Diode

3.4. Buzzer

A **buzzer** is a small yet efficient component to add sound features to our project/system. It is very small and compact 2-pin structure hence can be easily used on breadboard, and even on PCBs which makes this a widely used component in most electronic applications.

This buzzer can be used by simply powering it using a DC power supply ranging from 4V to 9V. A simple 9V battery can also be used, but it is recommended to use a regulated +5V or +6V DC supply. The buzzer is normally associated with a switching circuit to turn ON or turn OFF the buzzer at required time and require interval.

General description :

A buzzer or beeper is an audio signalling device, which may be mechanical, electromechanical, or piezoelectric. Typical uses of buzzers and beepers include alarm devices, timers and confirmation of user input such as a mouse click or key stroke. Buzzer is an integrated structure of electronic transducers, DC power supply, widely used in computers, printers, copiers, alarms, electronic toys, automotive electronic equipment, telephones, timers and other electronic products for sound devices. Active buzzer 5V Rated power can be directly connected to a continuous sound, this section dedicated sensor expansion module and the board in combination, can complete a simple circuit design, to "plug and play."

Product description :

A buzzer or beeper is an audio signalling device, which may be mechanical, electromechanical, or piezoelectric. Typical uses of buzzers and beepers include alarm devices, timers, and confirmation of user input such as a mouse click or keystroke. It generates consistent single tone sound just by applying D.C voltage. Using a suitably designed resonant system, this type can be used where large sound volumes are needed.

Features:

- Input supply: 5 VDC
- Current consumption: 9.0 mA max.
- Oscillating frequency: 3.0 ±0.5 KHz
- Sound Pressure Level: 85dB min.

Applications of Buzzer

- Alarming Circuits, where the user has to be alarmed about something
- Communication equipments
- Automobile electronics
- Portable equipments, due to its compact size.
- Confirmation of user input (ex: mouse click or keystroke)
- Electronic metronomes
- Sporting events
- Judging Panels
- Annunciator panels

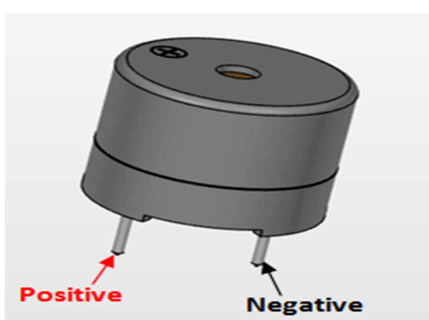

Fig. 3.4.1 Buzzer

3.5 Jumper Wires

Jumper wires are simply wires that have connector pins at each end, allowing them to be used to connect two points to each other without soldering. Jumper wires are typically used with breadboards and other prototyping tools in order to make it easy to change a circuit as needed. Fairly simple.

Jumper wires typically come in three versions: male-to-male, male-to-female and female-to-female. The difference between each is in the end point of the wire. Male ends have a pin protruding and can plug into things, while female ends do not and are used to plug things into. Male-to-male jumper wires are the most common and what you likely will use most often. When connecting two ports on a breadboard, a male-to-male wire is what you'll need.

In this project male-to-female jumper wires are used.

Fig. 3.5.1 Jumper Wires

4. WORKING

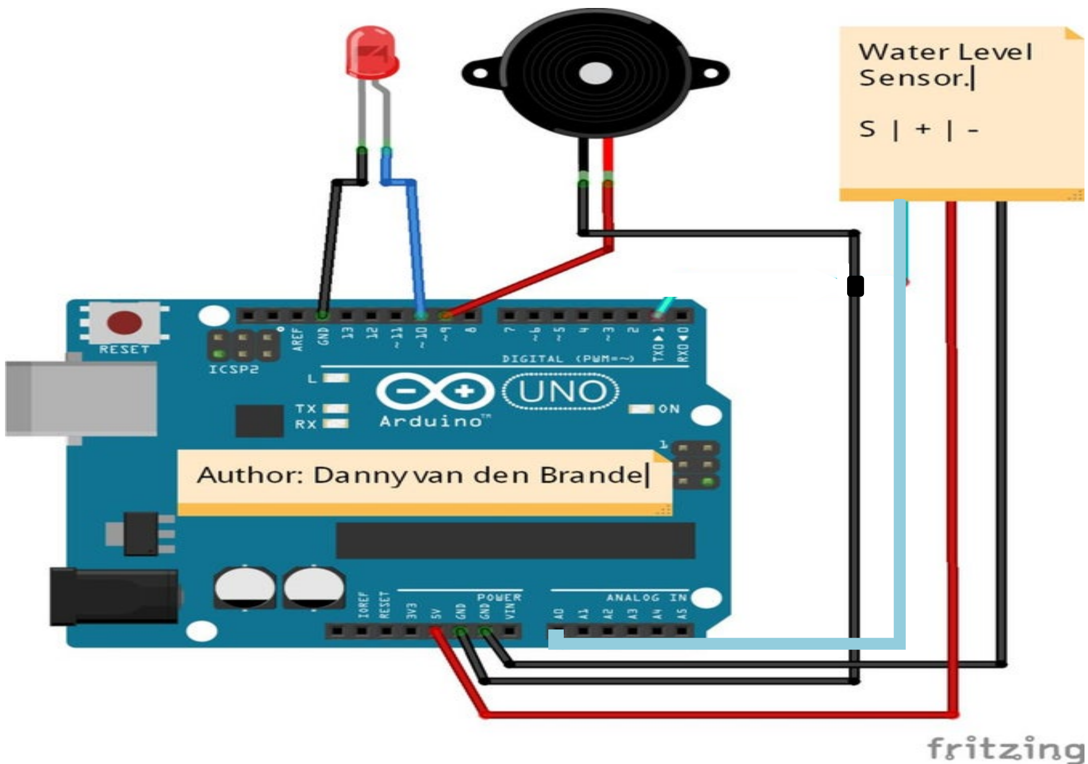

Fig 4.1 Circuit diagram

S pin of analog water sensor is connected to analog pin A0 of arduino board. Similarly +Vcc and –GND pins of water sensor are connected to +5v and GND pins of arduino uno. Positive and negative terminals of led are connected to 10 and GND pins of arduino respectively. Positive and negative terminals of buzzer are connected to arduino pins 9 and GND respectively. When the sensor detects water pin A0 goes high. According to the code led glows and buzzer rings while A0 pin goes high else led and buzzer pins goes low if A0 pin is low.

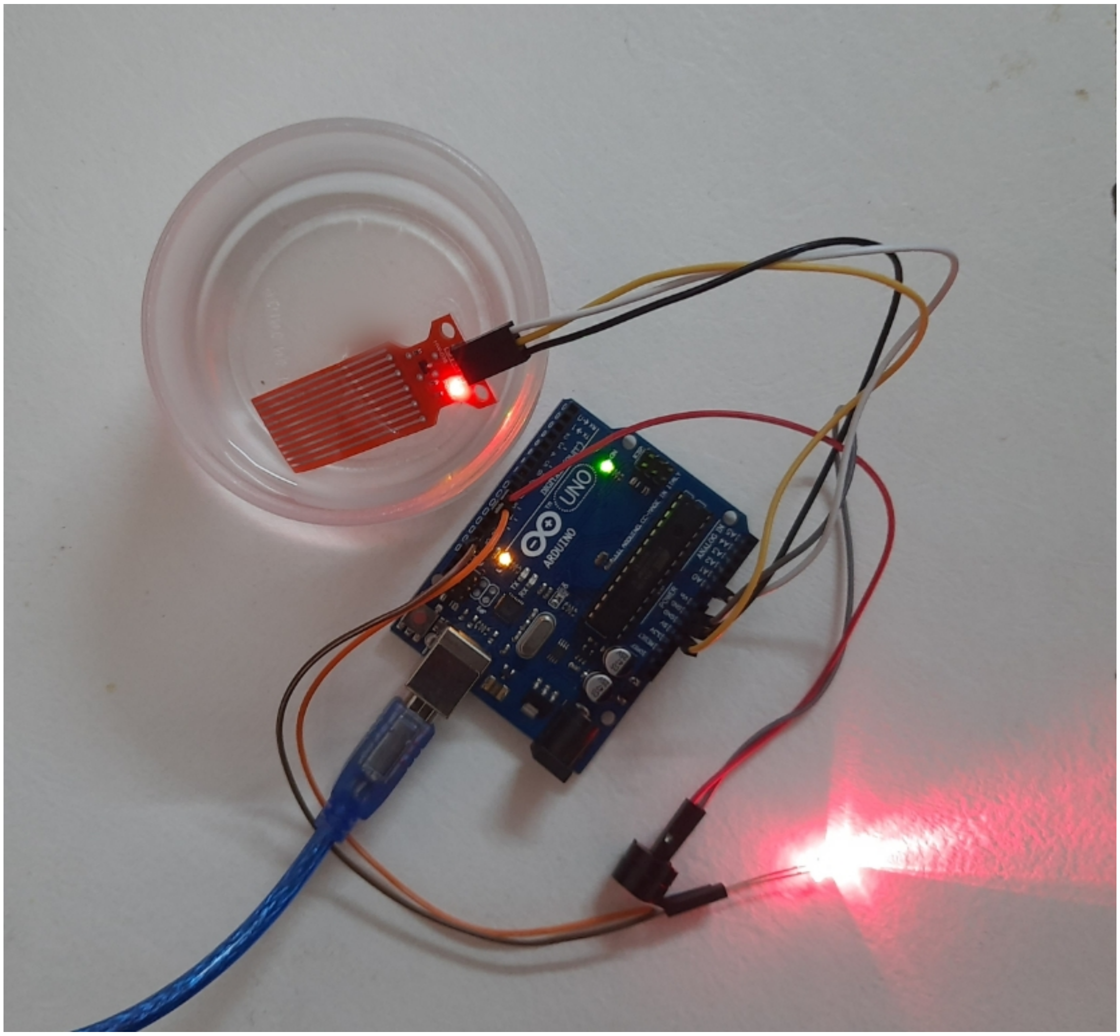

Fig.4.1.2 Arduino Flood alarm circuit

5. SOFTWARE USED

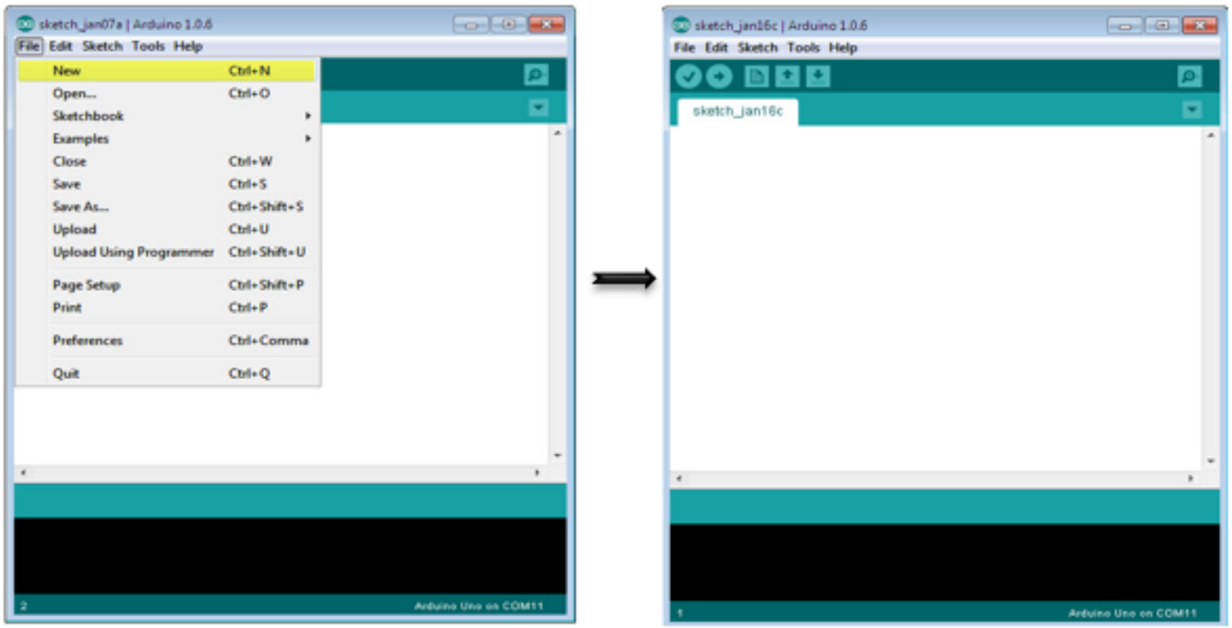

Fig.5.1.1 Software used

5.1 Introduction to Arduino IDE :

The Arduino Uno is programmed using the Arduino Software (IDE), Integrated Development Environment common to all our boards and running both online and offline.

The Arduino Software (IDE) allows writing programs and uploading them to arduino board.

1. If we have a reliable Internet connection, we can use the online IDE (Arduino Web Editor). It will allow us to save your sketches in the cloud, having them available from any device and backed up. We will always have the most up-to-date version of the IDE without the need to install updates or community generated libraries.

2. If we would rather work offline, we should use the latest version of the desktop IDE.

If we want to program your Arduino Uno while offline you need to install the Arduino Desktop IDE. The Uno is programmed using the Arduino Software (IDE), our Integrated Development Environment common to all our boards. Before we can move on, you must have installed the Arduino Software (IDE) on your PC. Connect Uno board with an A B USB cable; sometimes this cable is called a *USB printer cable.*

Arduino IDE is official software introduced by Arduino.cc that is mainly used for writing, compiling and uploading the code in the Arduino Device. Almost all Arduino modules are compatible with this software that is an open source and is readily available to install and start compiling the code on the go.

- Arduino IDE is open source software that is mainly used for writing and compiling the code into the Arduino Module.
- It is official Arduino software, making code compilation too easy that even a common person with no prior technical knowledge can get their feet wet with the learning process.
- It is easily available for operating systems like MAC, Windows, Linux and runs on the Java Platform that comes with inbuilt functions and commands that play a vital role for debugging, editing and compiling the code in the environment.
- A range of Arduino modules available including Arduino Uno, Arduino Mega, Arduino Leonardo, Arduino Micro and many more.
- Each of them contains a microcontroller on the board that is actually programmed and accepts the information in the form of code.
- The main code, also known as a sketch, created on the IDE platform will ultimately generate a Hex File which is then transferred and uploaded in the controller on the board.
- The IDE environment mainly contains two basic parts: Editor and Compiler where former is used for writing the required code and later is used for compiling and uploading the code into the given Arduino Module.
- This environment supports both C and C++ languages.

5.2 Arduino IDE :

We can download the Software from Arduino main website. The IDE environment is mainly distributed into three sections

- **1. Menu Bar**
- **2. Text Editor**
- **3. Output Pane**

The bar appearing on the top is called **Menu Bar** that comes with five different options as follow

- **File** – You can open a new window for writing the code or open an existing one. Following table shows the number of further subdivisions the file option is categorized into.

File	
New	This is used to open new text editor window to write your code
Open	Used for opening the existing written code
Open Recent	The option reserved for opening recently closed program
Sketchbook	It stores the list of codes you have written for your project
Examples	Default examples already stored in the IDE software
Close	Used for closing the main screen window of recent tab. If two tabs are open, it will ask you again as you aim to close the second tab
Save	It is used for saving the recent program
Save as	It will allow you to save the recent program in your desired folder
Page setup	Page setup is used for modifying the page with portrait and landscape options. Some default page options are already given from which you can select the page you intend to work on
Print	It is used for printing purpose and will send the command to the printer
Preferences	It is page with number of preferences you aim to setup for your text editor page
Quit	It will quit the whole software all at once

Fig.5.2.1 File subdivisions

The Output Pane will show the code compilation as you click the upload button. And at the end of compilation, it will show you the hex file it has generated for the recent sketch that will send to the Arduino Board for the specific task you aim to achieve.

- **Edit** – Used for copying and pasting the code with further modification for font
- **Sketch** – For compiling and programming
- **Tools** – Mainly used for testing projects. The Programmer section in this panel is used for burning a bootloader to the new microcontroller.
- **Help** – In case you are feeling skeptical about software, complete help is available from getting started to troubleshooting.

The **Six Buttons** appearing under the Menu tab are connected with the running program as follow.

Fig.5.2.2 Menu Bar

- The check mark appearing in the circular button is used to verify the code. Click this once you have written your code.
- The arrow key will upload and transfer the required code to the Arduino board.
- The dotted paper is used for creating a new file.
- The upward arrow is reserved for opening an existing Arduino project.
- The downward arrow is used to save the current running code.
- The button appearing on the top right corner is a **Serial Monitor** – A separate pop-up window that acts as an independent terminal and plays a vital role for sending and receiving the Serial Data. We can also go to the Tools panel and select Serial Monitor, or pressing Ctrl+Shift+M all at once will open it instantly. The Serial Monitor will actually help to debug the written Sketches where you can get a hold of how your program is operating. Arduino Module should be connected to computer by USB cable in order to activate the Serial Monitor.

The bottom of the main screen is described as an Output Pane that mainly highlights the compilation status of the running code: the memory used by the code, and errors occurred in the program. You need to fix those errors before you intend to upload the hex file into your Arduino Module.

More or less, Arduino C language works similar to the regular C language used for any embedded system microcontroller, however, there are some dedicated libraries used for calling and executing specific functions on the board.

5.3 How to select the Board :

In order to upload the sketch, you need to select the relevant board you are using and the ports for that operating system as shown in Fig.5.3.

- After correct selection of both Board and Serial Port, click the verify and then upload button. The sketch is written in the text editor and is then saved with the file extension .ino.

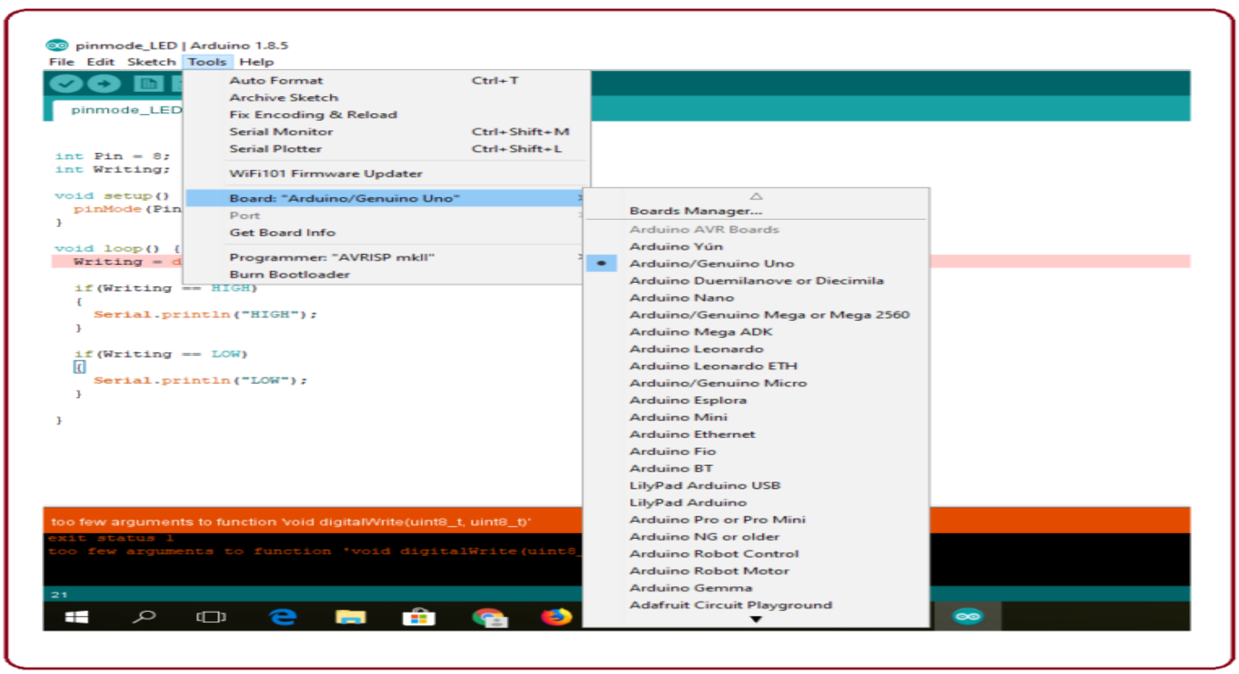

Fig.5.3.1 Arduino Board selection

6. ADVANTAGES

- Easy installation.
- Minimal maintaince.
- Saves money by using less electricity.
- Eco-Friendly.
- Easy to operate and fast response.
- Efficient design.

7. APPLICATIONS

- Most of the manmade floods occur with a pipe bursting, plumbing problems, or when a malfunctioning of water heater is present.
- Flood alarm is helpful for people in frequently flooded areas.

6. CONCLUSION

Floods can cause many problems when they occur and even take lives. They can affect people mentally and emotionally, causing long term mental health problems. Further more flooding destroys crops plants and other aspects of environment.

Flood monitoring is a clever and productive way to monitor floods and prevent damage when a disaster strikes, at any moment. They essentially pre-warn homeowners or residents of incoming floods.

The study is all about detecting the flood. A flood alarm can save a lot of trouble associated with flooding. We can take a proactive approach to both manmade floods and natural disasters. Because floods can happen at anytime, it’s better to be prepared for it.

7. FUTURE SCOPE

A battery comes with flood alarm, but it may last only for certain period of time. Hence they need to be checked regularly to be functional at all times. A possible means of power supply for the sensors and centralized control unit is via solar cells. The flood alert information's can be displayed on LED display boards for road users and for safety reasons could be placed at strategic locations. Such information's should be in real time and transmitted wirelessly from the measured location. The Flood Observatory System will be easy to install and maintained if it is powered by solar cells. The use of solar energy will also provide cheaper source of power to the entire system.

8. ANNEXURE

```
int WaterAlarm = A0;

int Buzzer = 9;

int Led = 10;

boolean val; //simply reads 1 or 0 as a value 1 when detecting water


// here i set up the tones, you can change them @ void loop.

int tones[] = {261, 277, 293, 311, 329, 349, 369, 392, 415, 440, 466, 493, 523 ,554};

//              1    2    3    4    5    6    7    8    9    10   11   12   13   14

// You can add more tones but i added 14. Just fill in what tone you would like to use, @ void loop you see " tone(Buzzer, tones[12]); " below,  digitalWrite(Buzzer, HIGH);

// here you can change the tones by filling in a number between 1 and 14


void setup() {

  Serial.begin(9600);

  pinMode (WaterAlarm, INPUT);

  pinMode (Buzzer, OUTPUT);

  pinMode (Led, OUTPUT);


}
```

```
void loop() {

  val = analogRead(WaterAlarm);
  if (val >= 1)
  {
  digitalWrite(Led, HIGH);
  digitalWrite(Buzzer, HIGH);
  tone(Buzzer, tones[6]);
  delay(200);
  digitalWrite(Led, LOW);
  digitalWrite(Buzzer, LOW);
  noTone(Buzzer);
  delay(200);
  digitalWrite(Led, HIGH);
  digitalWrite(Buzzer, HIGH);
  tone(Buzzer, tones[14]);
  delay(200);
  digitalWrite(Led, LOW);
  digitalWrite(Buzzer, LOW);
  noTone(Buzzer);
  delay(200);
  }
```

```
  else
  {
  digitalWrite(Led, LOW);
  digitalWrite(Buzzer, LOW);
  }
}
```

Reference

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2. https://www.covesmart.com/blog/protecting-your-home-what-are-flood-sensors-and-adv
3. www.tutorialspoint.com